

Loan Portfolio Dashboard

Power BI Data Analyst Project

1. Business Questions Prioritised

With a 50,000-account portfolio and six product lines, the dashboard was built around four questions leadership needs answered at a glance:

- **Is portfolio value concentrated where risk is lowest, or highest?** Page 1 shows 4-Wheeler drives 77.5% of disbursed value; Page 2 shows it also carries the lowest NPA rate (24.96%) of any product. Value and risk concentration point the same direction — a reassuring structural finding, not a coincidence worth leaving to chance.
- **Which levers actually move NPA risk, and which don't?** Product category and DPD-based restructuring status are strong differentiators (24.96%–64.13% NPA range by product; 28.37%–69.46% by restructure count). Employment type and branch, by contrast, barely move the needle (36.2%–38.6% NPA range across employment types; TAT and Collection Efficiency both sit in tight bands regardless of branch). Distinguishing real signals from noise keeps leadership from chasing the wrong fix.
- **Where is overdue money actually concentrated?** The aging waterfall on Page 5 shows accounts past 180 days hold 71.5% of all overdue value (NPR 107.17 Cr of NPR 149.97 Cr) despite being only 25% of overdue accounts by count — collections effort is best triaged toward preventing accounts from crossing into severe delinquency, not spread evenly.
- **Is the portfolio's reported NPA rate actually correct?** NPA_Flag only ever co-occurs with Loan_Status = "NPA", never "Active" — so NPA Rate had to be built on Active + NPA ("open") accounts as the denominator, not Active alone. Using Active alone overstates NPA Rate at 57.76% instead of the correct 36.61%, a 21-point error that would have misrepresented portfolio health to the Chairman.

2. DAX Measures I'm Proud Of

- **NPA Rate %** — $\text{DIVIDE}([\text{NPA Count}], [\text{Total Open Accounts}], 0)$, where Total Open Accounts = Active + NPA status. Correcting the denominator (see above) was the single most consequential modelling decision in the whole build.
- **Portfolio at Risk (PAR30)** — outstanding balance of DPD > 30 accounts as a share of total outstanding, calculated independently of formal NPA/NRB classification. It gives a purely behaviour-based early-warning signal (22.25%) that sits between the softer Overdue Ratio (10.24%) and the classification-based NPA Rate (36.61%), triangulating portfolio stress from three angles.
- **NPA Rate by Disbursement Vintage** — built on the existing Disbursement_Date relationship rather than a new column. It surfaces vintage seasoning bias cleanly: 2019 cohorts show 60–70%+ NPA rates versus ~21% for 2024 cohorts, which looks like dramatically improving credit quality but is largely an artefact of older loans having had more time to season into default. Flagging that caveat on the dashboard itself, rather than letting the trend line mislead, was a deliberate choice.

3. An Insight I Found Interesting

Restructured loans are, in this portfolio, a leading indicator of default rather than a cure. Accounts with zero restructures carry a 28.37% NPA rate; one restructure jumps that to 64.01%; two or more restructures reach 69.46% — and notably, 100% of twice-restructured accounts remain open (none have ever closed successfully). Restructuring is often framed as a relief mechanism that saves a loan, but the data here says the opposite: once a loan needs restructuring, it very rarely recovers. That's a concrete, actionable finding for the credit and risk team — restructured accounts likely warrant materially closer post-restructure monitoring, and restructuring policy itself may deserve a second look rather than being treated as a routine forbearance tool.